Adaptive Attacker Stress Test

Can a stealth attacker hide under the dead-zone, and does a second detection signal help?

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Headline result

The attacker has no profitable hiding place. Evasion and effectiveness are mutually exclusive: the poison rate low enough to slip under the detection threshold is the same poison rate at which the backdoor stops working. A second detection signal we proposed in advance turned out to be unnecessary and slightly harmful, so it is reported as a measured negative result rather than adopted.

Why we ran this

The improvement adopted in the previous study (a gentler trust gate at beta = 1.0 plus a suspicion dead-zone at tau = 2.0) cut the honest false-positive rate from 20.5% to 0.3%. But a dead-zone is a threshold, and a threshold invites gaming. An attacker does not have to beat the behavioral probe; it only has to keep its suspicion below tau and it is never gated at all. Since we introduced that possibility ourselves, we tested it, framed as a security question: what is the largest backdoor an attacker can implant while staying undetected?

What was tested

- Adaptive stealth attacker: drops the model-replacement scaling entirely and sweeps its poison rate down from 40% to 1%, trading backdoor strength for invisibility.
- D2 (current best): behavioral probe signal only.
- D4 (proposed fix): max(behavioral probe, update-direction anomaly). The direction signal is the robust MAD z-score of each client cosine distance from the coordinate-wise median update direction, and it does not require activating the trigger.

Both use the same gate, dead-zone, and coordinate-wise median aggregation, so the only difference is the added signal. Everything runs across seeds 42, 7, 123.

Results

Poison rate	Defense	Backdoor Lift	Attacker detected	Honest FP	Attacker probe susp.	Attacker direction susp.
1%	**No defense**	**0.0051 +/- 0.0013**	**n/a**	**n/a**	**n/a**	**n/a**
1%	**D2 probe only**	**-0.0034 +/- 0.0108**	**26.4%**	**3.5%**	**3.26 +/- 0.85**	**0.78 +/- 0.30**
1%	**D4 probe + direction**	**-0.0093 +/- 0.0165**	**33.3%**	**6.9%**	**3.42 +/- 0.61**	**0.76 +/- 0.37**
5%	No defense	0.0354 +/- 0.0027	n/a	n/a	n/a	n/a
5%	D2 probe only	-0.0216 +/- 0.0203	100.0%	0.7%	7.35 +/- 0.99	2.22 +/- 0.43
5%	D4 probe + direction	-0.0184 +/- 0.0189	100.0%	1.4%	7.77 +/- 1.05	2.25 +/- 0.42
10%	No defense	0.0564 +/- 0.0046	n/a	n/a	n/a	n/a
10%	D2 probe only	-0.0264 +/- 0.0203	100.0%	1.0%	12.00 +/- 2.45	6.99 +/- 0.74
10%	D4 probe + direction	-0.0251 +/- 0.0192	100.0%	1.4%	13.18 +/- 2.91	7.10 +/- 0.64
20%	No defense	0.0877 +/- 0.0095	n/a	n/a	n/a	n/a
20%	D2 probe only	-0.0266 +/- 0.0195	100.0%	1.0%	20.75 +/- 3.49	20.03 +/- 1.86
20%	D4 probe + direction	-0.0275 +/- 0.0169	100.0%	1.0%	19.93 +/- 2.84	18.44 +/- 1.59
40%	No defense	0.1369 +/- 0.0311	n/a	n/a	n/a	n/a
40%	D2 probe only	-0.0266 +/- 0.0186	100.0%	0.3%	29.56 +/- 5.77	37.61 +/- 2.74
40%	D4 probe + direction	-0.0288 +/- 0.0168	100.0%	1.0%	29.62 +/- 4.71	39.54 +/- 4.23

Table 1. Adaptive stealth attacker swept from 40% poison down to 1%, mean +/- standard deviation over three seeds. The dead-zone is tau = 2.0; a suspicion score below it means the client is never gated. The 1% rows are highlighted because that is the only setting where detection begins to fail.

Verdict 1: the dead-zone holds, and the evasion frontier is unprofitable

From 5% poison upward the attacker probe suspicion is 7.35 or higher, more than three times the threshold, and it is detected in 100% of client-rounds. Detection only starts to fail at 1% poison, where it drops to 26.4%. But at that setting the undefended backdoor lift is +0.0051, which is indistinguishable from zero. In other words, the poison rate the attacker needs in order to hide is the same poison rate at which the backdoor stops doing anything. This is a stronger statement than "we did not find an evasion": we mapped the attacker operating curve and showed that its best available move is not worth taking.

Verdict 2: the proposed second signal is rejected

The hypothesis was that update-direction anomaly would catch a subtle attacker that the behavioral probe misses, because it does not depend on activating the trigger. The measurement says the opposite. At exactly the stealthy settings where it was supposed to help most, the direction signal is the weaker of the two: 0.76 at 1% poison and 2.25 at 5%, against 3.26 and 7.35 for the behavioral probe. It only overtakes the probe at 40% poison, where the probe already catches everything. It also roughly doubles the honest false-positive rate at 1% poison (6.9% against 3.5%) and produces no improvement in backdoor lift. Adding it would make the defense more complex and slightly worse, so it is not adopted.

The reason is clear in hindsight. The behavioral probe measures the thing we care about directly, by activating the backdoor behavior. Update direction is a proxy: it detects that a client optimized something different, but at low poison rates the backdoor is such a small part of the client objective that its update still points essentially where the honest updates point. The proxy degrades faster than the direct measurement, which is the reverse of the intuition that motivated it.

Figure

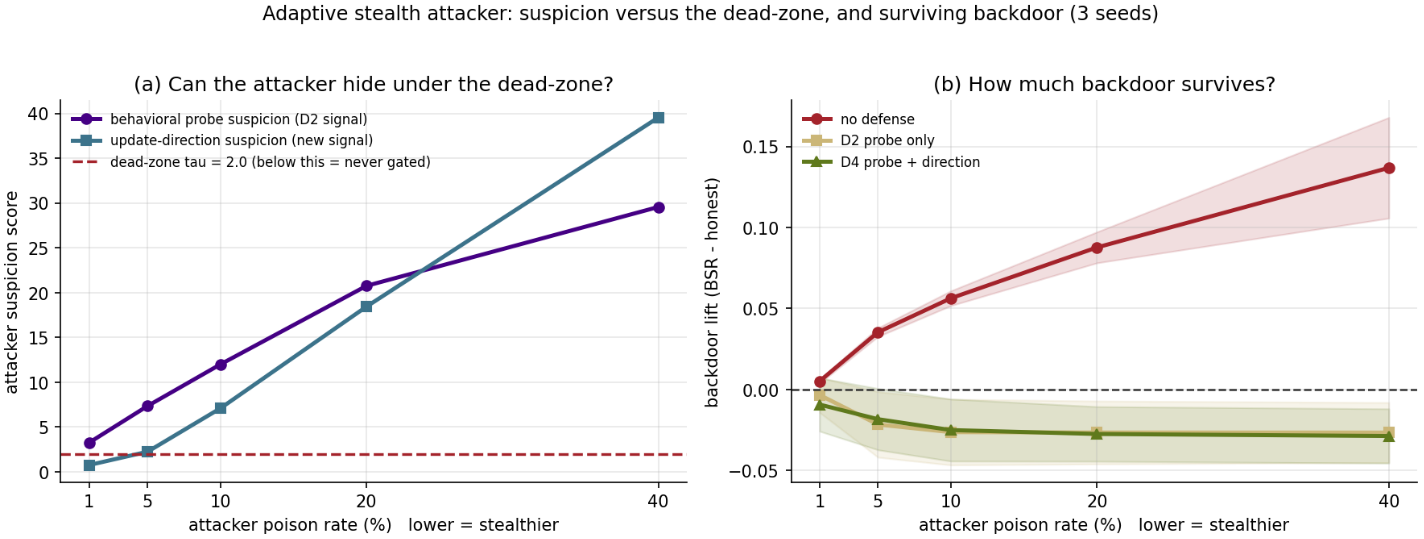

Figure 1. Adaptive stealth attacker. Panel (a): the attacker two suspicion scores against the dead-zone threshold (red dashed line) as the poison rate falls. Panel (b): the surviving backdoor lift with no defense, with D2, and with D4. Bands are +/- 1 standard deviation over three seeds.

Reading the figure

Panel (a) shows the attacker becoming harder to see as it gets stealthier, with the purple probe line falling toward the threshold at 1% poison. Panel (b) shows why that invisibility is worthless: the red no-defense curve, which is the damage the attacker would do if the defense were absent entirely, collapses to the zero line at exactly the same point. The gold and green defended curves sit at or below zero across the whole sweep, and are indistinguishable from each other, which is the visual form of the conclusion that the second signal adds nothing.

Scope and Limitations

What this establishes

D2 resists the natural adaptive strategy: poison less and stop scaling in order to look ordinary. Across the whole stealth range the attacker either is caught outright or gains no measurable backdoor.

What this does not establish

This is not a general claim of adaptive-attacker robustness, and we do not present it as one. It tests one adaptive strategy. It does not test a stronger attacker that explicitly optimizes against the defense, for example one that adds a penalty for probe-slice error to its local training loss, or that solves a constrained problem maximizing backdoor effect subject to keeping suspicion below tau. That attacker is the real remaining frontier and is the single most valuable next experiment.

Also unchanged

- The base detector is still only moderately strong, with honest spoofing recall around 0.55, which is why backdoor lift rather than raw accuracy remains the primary metric.
- The setting is still IID with ten clients and two attackers. Non-IID client data would widen the honest trust spread and is the natural next deeper experiment.
- The dead-zone threshold tau = 2.0 was chosen by reasoning about the MAD scale rather than swept.

Reproducibility

Every number and the figure are produced by adaptive_attacker.ipynb and exported to results/ as adaptive_attacker_sweep.csv and fig_adaptive_attacker.png. This PDF is assembled by build_adaptive_pdf.py, which reads those files directly, so the report cannot drift from the experiment.